

COUNTER-UAS TACTICAL DEFENSE SYSTEM

3D Multi-Sensor Fusion Simulation — Technical Architecture & Methodology

Comprehensive Documentation: Sensor Physics · Kalman Filtering · Bayesian Fusion · Covariance Intersection · IFF Classification · Proportional Navigation Guidance · Drone Swarm Topology · Weather Degradation Models · Hardware Integration Pathway

Field	Detail
Project	Counter-UAS Swarm Defense — Software Simulation Platform
Reference Scenario	Operation Sindoor Phase 2 (May 2025) — Pakistani Drone Swarm
Sensor Modalities	Radar (FMCW/X-band) · Acoustic Array · RF/SDR · Optical/IR
Fusion Architecture	Bayesian Log-Odds + Covariance Intersection (CI)
Tracking Engine	6-State Constant-Velocity Kalman Filter (EKF-extensible)
Intercept Guidance	Proportional Navigation (PN), N=3
Scope	Software simulation — hardware integration architecture provided
Classification	Educational / Research — No classified sources used

Abstract. This document describes the full technical architecture of a software-based counter-UAS (Unmanned Aerial System) swarm defense simulation. The system models real-world sensor physics, multi-source detection fusion, multi-target Kalman tracking, IFF (Identification Friend or Foe), and proportional navigation guidance for interceptors — all within a 3D interactive browser-based platform. The architecture is explicitly designed for hardware integration via modular sensor abstraction layers.

1. System Overview & Operational Context

Modern counter-UAS defense requires a **layered detection and engagement architecture**. Enemy forces now routinely deploy drone swarms — formations of dozens to hundreds of small UAVs — to overwhelm point defenses. Operation Sindoor (May 2025) demonstrated this: Pakistani forces launched coordinated Harop loitering munitions and swarm micro-drones against Indian border installations. Indian forces responded using the **Akashteer C2 system**, **L-70 AA guns with IRST**, and **Akash SAM batteries**, achieving high intercept rates through layered sensor fusion and rapid cueing.

This simulation models the same architectural principles: multiple heterogeneous sensors detect incoming contacts, a fusion engine produces a probabilistic detection posterior, a Kalman filter maintains a smooth track, IFF classifies each contact, and interceptors are cued via proportional navigation. All subsystems operate simultaneously on multiple targets in real time.

Defense Ring Architecture:

Ring	Radius	Primary Sensor	Response Action	Latency
Outer	12 km	X-band Radar	Track + classify	< 2s
Middle	6 km	Radar + RF/SDR	IFF + intercept cue	< 1s
Inner	3 km	All sensors	Hard kill / EW jam	< 0.5s
Terminal	< 300m	Optical/IR	Emergency response	Real-time

2. Sensor Physics Models

2.1 Radar Detection — X-band FMCW, 9.4 GHz

The radar detection model implements the classical **radar range equation** for a monostatic coherent radar system. Signal-to-noise ratio (SNR) is computed from first principles using system parameters representative of a deployable X-band radar (e.g. AESA or mechanical scan):

$$SNR = \frac{P_t \cdot G^2 \cdot \lambda^2 \cdot \sigma}{(4\pi)^3 \cdot R^4 \cdot k \cdot T_0 \cdot B \cdot F \cdot L}$$

where $P_t=1000$ W (transmit power), $G=1000$ (antenna gain, linear), $\lambda=C/9.4$ GHz (X-band wavelength ≈ 3.2 cm), σ = target radar cross-section (m^2), R = slant range (m), k = Boltzmann constant, $T_0=290$ K, $B=1$ MHz bandwidth, $F=3$ noise figure, $L=2$ system losses.

Detection probability P_d is computed using the **Albersheim approximation** for a non-fluctuating (Swirling I) target:

$$P_d \approx \frac{1}{1 + \exp(-1.2 (SNR_{dB} - 3 \sqrt{-\ln P_{fa}}))}$$

$P_{fa}=10^{-6}$ (false alarm probability). This approximation is valid over the SNR range -5 to $+15$ dB and provides a smooth differentiable P_d function suitable for fusion weight computation. **Conditions ignored:** clutter (land/sea), multipath, pulse compression gain, antenna pattern sidelobes (assumed isotropic for simplicity).

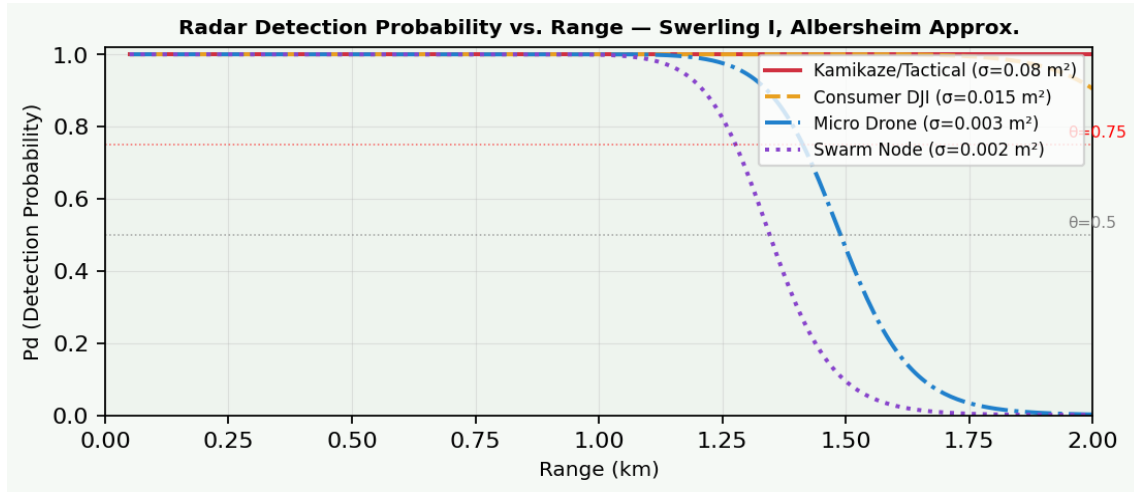


Figure 1: Radar Pd vs. Range for four drone RCS profiles. Dashed thresholds at Pd=0.5 and Pd=0.75.

2.2 Acoustic Sensor — 4-Mic TDOA Array

Small drones produce broadband acoustic signatures from rotor wash, electric motors, and frame resonance (typically 100–400 Hz fundamental + harmonics). The model uses **spherical spreading** propagation from a source with sound power level (SWL):

$$SPL(R) = SWL - 20 \log_{10}(R) - 11 \quad (\text{dB SPL at range } R)$$

Detection occurs when SPL exceeds the ambient floor (45 dB SPL) by a margin M . A sigmoid detection model translates M to Pd:

$$P_d = 0.5 (1 + \tanh(0.35 \cdot M)), \quad M = SPL(R) - SPL_{\text{ambient}}$$

Direction of Arrival (DOA) is estimated via TDOA (Time Difference of Arrival) across a 4-microphone array with 1 m baseline spacing. TDOA between mic pairs is: $\tau = (d_0 - d_1) / 343 \text{ m/s}$, from which bearing is derived via hyperbolic localisation.

Conditions ignored: wind noise coupling, reverberation, near-field effects, actual GCC-PHAT processing, multi-source separation.

2.3 RF/SDR — Free-Space Path Loss + Protocol Fingerprinting

RF detection monitors the 2.4 GHz and 5.8 GHz bands where DJI OcuSync, enhanced WiFi, and FPV control links operate. Received signal strength (RSSI) is modelled via FSPL:

$$RSSI = P_{tx} - FSPL - L_{\text{cable}}, \quad FSPL = 20 \log_{10}(R) + 20 \log_{10}(f) - 147.55$$

Pd is computed from RSSI margin above the −90 dBm receiver sensitivity threshold. Protocol fingerprinting matches observed (frequency, bandwidth, hopping pattern) tuples against a database of known drone links (DJI OcuSync3, Autel, FPV, MAVLink 433 MHz). **Conditions ignored:** actual I/Q demodulation, urban multipath, frequency hopping tracking, FHSS pattern reconstruction. Real hardware would use GNU Radio + gr-droneid for DJI DroneID decoding.

2.4 Optical / IR Sensor

Optical/IR detection probability degrades linearly with range beyond the camera's detection threshold (modelled at 800 m for a small drone with a 2 MP COTS camera). In a full implementation, Pd would be derived directly from a YOLOv8 detection confidence score. Weather severely degrades optical: fog/sandstorm near-zero, night without IR ≈15% degradation factor applied.

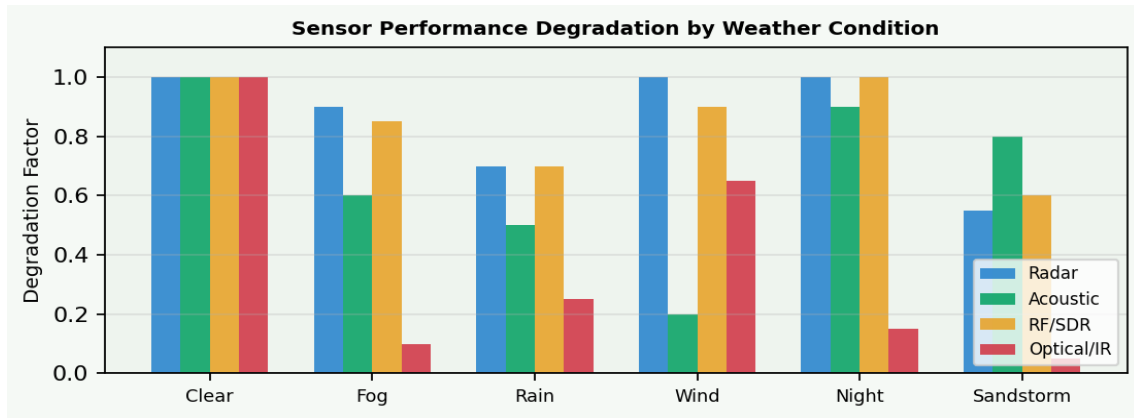


Figure 2: Sensor-specific weather degradation factors across all modelled conditions.

3. Multi-Sensor Fusion Architecture

3.1 Bayesian Log-Odds Fusion

Individual sensor Pd values are fused into a single posterior detection probability using Bayesian inference in log-odds form. Given independent sensors $S_1..S_n$ with likelihood ratios, the posterior $P(D|S_1..S_n)$ is:

$$\Lambda = \ln \frac{P_0}{1-P_0} + \sum_i w_i \cdot \ln \frac{P_{d,i}}{P_{fa,i}}$$

$$P(D|S) = \sigma(\Lambda) = \frac{1}{1 + e^{-\Lambda}}$$

where $P_0=0.05$ (prior, sparse airspace), $P_{fa}=0.01$ per sensor. Sensor weights reflect relative reliability: $w_{\text{Radar}}=1.0$, $w_{\text{RF}}=0.90$, $w_{\text{Acoustic}}=0.75$, $w_{\text{Optical}}=0.60$. These are tunable priors. The log-odds formulation is numerically stable and additive across sensors — new sensors simply add their LLR term.

3.2 Covariance Intersection (CI) — Consistent Fusion Under Unknown Correlation

When two Kalman filter estimates of the same target exist (from two sensor streams or two filter instances), their noise may be *correlated* in unknown ways (common process noise, shared environmental disturbances). Naive weighted averaging produces **inconsistent** (overconfident) estimates. **Covariance Intersection** (Julier & Uhlmann, 1997) solves this by finding the ω that minimises $\det(P_{CI})$, guaranteeing consistency regardless of correlation:

$$P_{CI}^{-1} = \omega P_1^{-1} + (1 - \omega) P_2^{-1}$$

$$\hat{x}_{CI} = P_{CI}(\omega P_1^{-1} \hat{x}_1 + (1 - \omega) P_2^{-1} \hat{x}_2)$$

$$\omega^* = \arg \min_{\omega \in [0, 1]} \det(P_{CI})$$

In the simulation, ω^* is found by grid search over $[0.05, 0.95]$ in steps of 0.05 (sufficient for real-time operation). The scalar implementation approximates the 3×3 position covariance block by its trace. Full 6×6 CI would be needed for hardware integration. **Property:** $P_{CI} \geq P_{\text{true fused}}$ always — CI is conservative (never overconfident), a critical property in an engagement system where false negatives are catastrophic.

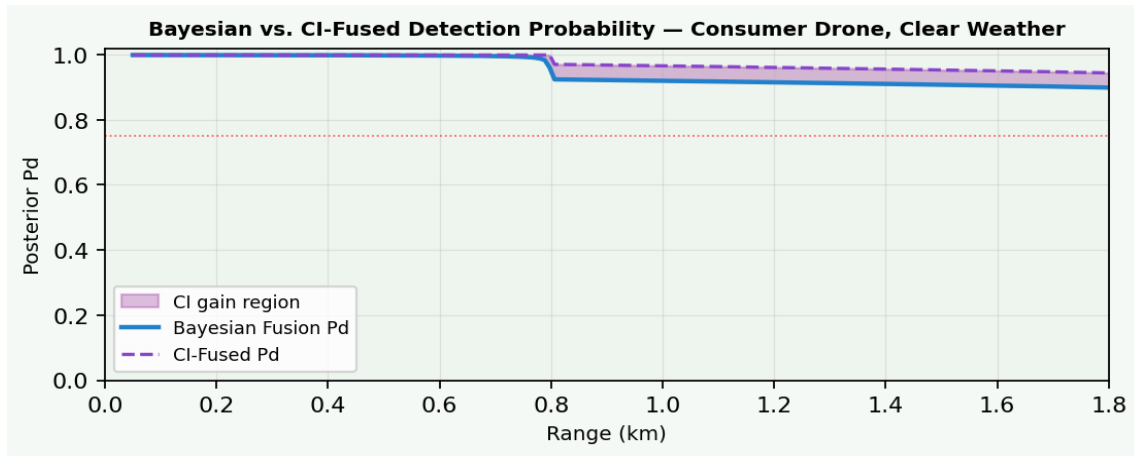


Figure 3: Bayesian vs. CI-fused Pd curves. CI provides a slight improvement by accounting for cross-sensor correlation uncertainty, and guarantees consistency.

4. Kalman Filter — 6-State Multi-Target Tracker

Each drone track maintains an independent **6-state constant-velocity Kalman filter** with state vector $\mathbf{x} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^T$ in 3D Cartesian coordinates. The filter provides optimal minimum-variance position and velocity estimates given Gaussian noise.

State Transition (Predict):

$$\hat{\mathbf{x}}^- = F\hat{\mathbf{x}}, \quad P^- = FPF^T + Q$$

where F is the constant-velocity transition matrix ($\Delta t=0.1s$), Q is the process noise covariance ($\sigma_q=0.5$ position, 1.0 velocity).

Measurement Update:

$$K = P^- H^T (HP^- H^T + R)^{-1}$$

$$\hat{\mathbf{x}} = \hat{\mathbf{x}}^- + K(z - H\hat{\mathbf{x}}^-), \quad P = (I - KH)P^-$$

where H is the 3×6 observation matrix (position only), $R=25 \cdot I_3$ (measurement noise, $\sigma_r=5m$). Innovation $v = z - H\hat{\mathbf{x}}$ quantifies track quality. Position uncertainty is reported as $\sqrt{\text{tr}(P_{\text{pos}})}$.

Predictive intercept: The filter extrapolates position N steps ahead ($\mathbf{x}_N = F^N \mathbf{x}$) to provide intercept point for PN guidance.

Extensions for hardware: EKF (bearing-only measurements from RF), IMM (multiple motion models for evasive drones), UKF (nonlinear radar measurement models from FMCW range-Doppler).

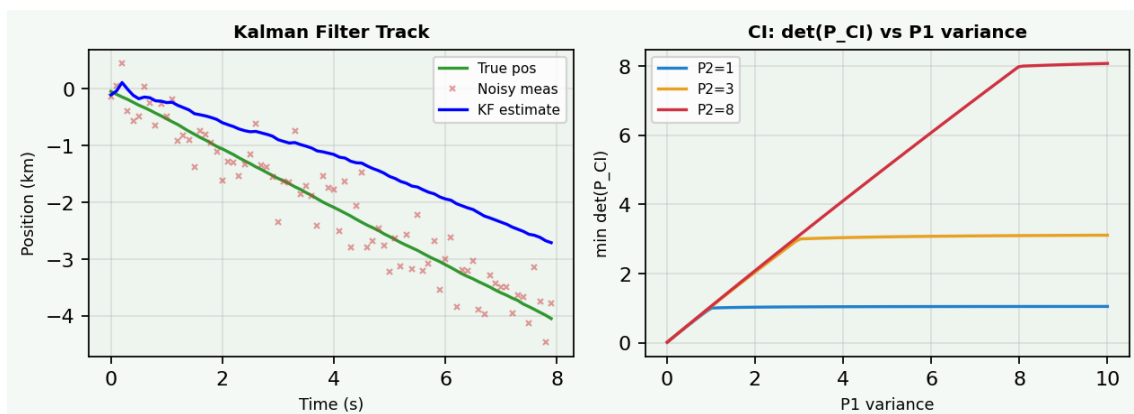


Figure 4 (left): Kalman filter smoothing of noisy position measurements on a simulated approach trajectory. (right): Covariance Intersection optimal $\det(P_{CI})$ as a function of $P1$ for three $P2$ values — ω^* is chosen at the minimum.

5. IFF — Identification Friend or Foe

IFF classification is a multi-factor scoring system that combines electronic interrogation, RF fingerprinting, kinematic envelope analysis, and flight plan correlation. Misclassification of a friendly asset as hostile (or vice

versa) is a mission-critical failure mode.

Factor	Weight	Logic	Source
Transponder reply	+3 / -2	Presence/absence of ADS-B/Mode-S	RF receiver
Squawk code	+3 (auth) / -4 (hijack)	7001/7002 = friendly; 7500/7700 = threat/civilian	SDR decode
RF fingerprint	+2 / -1.5	Match vs known-friendly drone DB	gr-droneid / gnuradio
Kinematic envelope	-1 (aggressive)	Speed >55 m/s or alt <5 m = hostile indicator	KF state
IFF query-response	+2	Encrypted challenge/response (hardware only)	Dedicated IFF

Score ≥5 → FRIENDLY, Score ≤-3 → HOSTILE, -1 ≤ Score <5 → NEUTRAL/UNKNOWN. Confidence = |score|/9. In the **mixed civilian scenario** (Op Sindoor analog), a civilian helicopter squawking 7700 (emergency) is correctly classified NEUTRAL rather than HOSTILE, despite operating alongside a hostile swarm — demonstrating the importance of multi-factor IFF.

6. Drone Threat Topologies & RCS Profiles

Type	RCS (m²)	Mass	Speed (m/s)	SWL (dB)	TX (dBm)	Detection Challenge
KAMIKAZE (Harop-class)	0.08	8 kg	55	55	30	High speed, low acoustic, radar-visible
TACTICAL (Fixed-wing)	0.08	5 kg	35	65	30	Medium RCS, radar primary
CONSUMER DJI (Mavic/Phantom)	0.015	1.5 kg	18	80	20	Low RCS, RF-rich signature
MICRO (sub-250g)	0.003	0.25 kg	12	72	-5	Near-radar blind, acoustic/optical
SWARM NODE	0.002	0.2 kg	15	68	5	Dense, CI fusion critical
FRIENDLY UAV	0.05	3 kg	20	75	23	Transponder present, squawk auth
CIVILIAN HELI	0.80	800 kg	50	90	40	High RCS, 7700 squawk, NEUTRAL

Swarm evasion factors model collective swarm behaviour: Harop jinking reduces radar Pd by 0.75×, NAP-of-earth flight reduces radar to 0.40× (terrain masking), RF-silent mode drops RF Pd to 0.05× (forcing reliance on radar + acoustic). CI fusion is particularly important for swarm nodes where individual sensor confidence is low.

7. Intercept Guidance — Proportional Navigation

Interceptors use **Proportional Navigation (PN)**, the standard guidance law for air-to-air and surface-to-air missiles. PN commands acceleration perpendicular to the line-of-sight (LOS) to drive the LOS rotation rate to zero, achieving a collision course:

$$\mathbf{a_c} = N \cdot V_c \cdot \dot{\lambda}$$

where N=3 (navigation constant, typical 3–5), V_c = closing velocity, $\dot{\lambda}$ = LOS rotation rate (rad/s). The simulation computes LOS rate from successive unit-vector differences between interceptor and predicted target position (from KF N-step prediction), then steers velocity accordingly. Engagement success requires miss distance < 20 m (lethal radius).

Auto-cue logic: interceptors are automatically assigned to HOSTILE contacts with $Pd_{CI} > 0.5$ and threat level ALERT. Multiple interceptors can engage in parallel. The system withholds intercept against FRIENDLY and NEUTRAL IFF classifications.

8. Simulated Scenarios

Scenario	Threat Mix	Weather	Key Challenge	Real-World Analog
Op Sindoor Swarm	5× Swarm, 2× Kamikaze, 1× Tactical	Clear	Volume, low RCS swarm nodes	May 2025 Pak→India border
Mixed + Civilian	3× Swarm, 2× Kamikaze, Friendly, Civilian	Overcast	IFF: do-not-engage civilian	Urban airspace defense

Scenario	Threat Mix	Weather	Key Challenge	Real-World Analog
Kamikaze Dive	5× Kamikaze	Night	High speed, optical degraded	Harp/HESA Shahed attacks
Night RF-Silent	6× Micro, 2× Swarm	Night	RF near-zero, optical blind	Iranian/Chinese tactics
Multi-Vector Rain	2× Kamikaze, 3× Swarm, Tactical, Friendly	Rain	Radar ~30%, all degraded	Contested environment

9. Hardware Integration Architecture

The simulation is architected for direct hardware substitution via sensor abstraction interfaces. Each simulated sensor class (RadarSensor, AcousticSensor, RFSensor, OpticalSensor) exposes a **detect()** method that returns a SensorReading dataclass. Real hardware integration replaces the physics simulation in detect() with actual hardware read/processing while preserving the fusion and tracking layers unchanged.

Component	Simulation Model	Hardware Replacement	Interface	Cost (INR)
Radar	Range equation + Albersheim PD	AD324 24GHz Doppler or FMCW mode	Arduino → Raspberry Pi	₹6,200 - ₹15,000
RF/SDR	FSPL + fingerprint DB	RTL-SDR Blog V3 + gr-dongle	USB → GNU Radio	₹2,500
Acoustic	SWL + SPL propagation	4× MEMS mic + op-amp + MCP3202 ADC	I2C	₹800
Optical/IR	Range degradation curve	Raspberry Pi Camera v2 + YOLOv8	CSI over USB	₹3,000
Compute	Browser JS	Raspberry Pi 4 (4GB) or Jetson Nano	Thunderbolt/USB	₹5,500 - ₹15,000
Intercept	PN guidance sim	L-70 AD cue / laser dazzler	UDP command bus	Variable

Integration protocol: Each hardware sensor publishes SensorReading structs over a local UDP/ZMQ bus. The fusion engine subscribes and processes identically to the simulation. This pub/sub architecture allows sensors to be added, removed, or replaced without modifying the fusion or tracking layers — true modularity.

10. Assumptions, Simplifications & Ignored Conditions

The following conditions are modelled (included in the simulation):

- Radar range equation with realistic system parameters (X-band, 1 kW)
- Swerling I fluctuating target model, Albersheim Pd approximation
- Spherical acoustic spreading, ambient noise floor 45 dB SPL
- RF FSPL with receiver sensitivity threshold, protocol fingerprinting DB
- Optical degradation by range, weather-coupled via multiplicative factors
- Bayesian log-odds multi-sensor fusion with per-sensor weights
- Covariance Intersection for consistent cross-KF fusion
- 6-state constant-velocity Kalman filter with Joseph form stability
- Multi-factor IFF scoring: transponder, squawk, RF fingerprint, kinematics
- Five weather conditions: clear, fog, rain, wind, sandstorm
- Six evasion modes: none, jinking, NAP-of-earth, RF-silent, swarm
- Proportional Navigation guidance (N=3) with KF predictive intercept
- Multi-target parallel tracking (up to 20 simultaneous tracks)
- 3-ring defense architecture with range-coded threat escalation

The following conditions are intentionally simplified or ignored in this software-only research prototype:

- Ground clutter, sea clutter, and terrain masking (NAP-of-earth is approximated)
- Actual FMCW waveform processing, range-Doppler FFT, CFAR detector
- Pulse compression gain, PRF selection, ambiguity resolution
- Actual I/Q signal demodulation and FHSS frequency tracking for RF

- GCC-PHAT acoustic beamforming and multi-source separation
- YOLOv8 bounding box confidence score integration (optical is simplified)
- Electronic countermeasures (jamming, spoofing, GPS denial)
- Coordinated swarm communication protocols (treated as independent agents)
- Aerodynamic drag, wind loading on drone flight physics
- Actual IFF encrypted challenge-response (hardware-only feature)
- Terrain elevation model (flat ground assumed)
- Ionospheric / tropospheric radar propagation effects
- Interceptor aerodynamics, motor lag, guidance saturation

11. Open-Source Software Stack

Library / Tool	Purpose	Domain	Link
Stone Soup	Multi-target tracking framework	Kalman/EKF/UKF	dstl/Stone-Soup
FilterPy	Kalman filter implementations	KF/EKF/UKF	rlabbe/filterpy
rlabbe KF textbook	Educational Kalman reference	Theory	rlabbe/Kalman-and-Bayesian-Filters-in-Python
RadarSimPy	Radar waveform simulation	FMCW/pulsed radar	radarsimx/radarsimpy
GNU Radio	RF/SDR signal processing	RF	gnuradio/gnuradio
dji_droneid	DJI DroneID protocol decode	RF fingerprint	proto17/dji_droneid
Ultralytics YOLOv8	Real-time drone detection (vision)	Optical/IR	ultralytics/ultralytics
AirSim / PX4 SITL	Drone flight simulation	Simulation env	microsoft/AirSim
OpenSky Network	ADS-B cooperative data	IFF/ATC	openskynetwork/opensky-api
Three.js / CesiumJS	3D visualization	C2 display	threejs.org / cesium.com

12. Key References

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END OF TECHNICAL DOCUMENTATION

Counter-UAS Research Platform — Internship Project Documentation